Term Project Checkpoint B

Machine Learning-Enhanced Adaptive Momentum ETF

Historical Data Preparation and Preliminary Findings

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Abstract

This checkpoint prepares historical market data for a proposed machine learning-enhanced momentum ETF. I collected dividend-adjusted daily price histories for SPY, XLF, XLV, IEF, GLD, and SHY and aligned them over November 2004 through October 2026. The common sample contains 5,518 daily prices. Preliminary results show substantial differences in volatility and drawdowns and high correlations among the equity ETFs. I also checked whether momentum and trend signals are available and compared three lookback windows. These are data-preparation findings, not strategy returns. Next steps are walk-forward portfolio backtesting, transaction costs, and a machine learning risk-management comparison.

Keywords: ETF, momentum, Monte Carlo simulation, adjusted returns, historical volatility, portfolio risk.

1. Introduction

The goal of this project is to test whether machine learning-based risk management improves the risk-adjusted performance of a transparent, rule-based momentum investment strategy. Checkpoint A defined the proposed fund, its investment philosophy, and its trading restrictions. Checkpoint B focuses on gathering the historical data needed to evaluate the same strategy rather than changing its rules.

The proposed fund is intended for investors who want a systematic alternative to discretionary stock selection. My main question remains: does an ML-based market-risk signal add value beyond a simpler monthly momentum approach, after accounting for trading and management costs? Before answering that question, I need a consistent history of all six instruments, reliable return calculations, and a clear plan for when information becomes available.

2. Literature Review

Portfolio theory and trend-following research motivate different parts of the design. Markowitz (1952) shows why dependence between assets matters for portfolio risk. Moskowitz, Ooi, and Pedersen (2012) document time-series momentum across several markets, while Faber (2013) examines rules-based allocation across asset classes. Together, these studies support starting with a simple, transparent strategy but do not establish that this particular ETF portfolio will outperform its benchmark.

Moreira and Muir (2017) study adjustments to market exposure when volatility changes. Gu, Kelly, and Xiu (2020) demonstrate how machine learning can model nonlinear predictive relationships in financial data. These studies motivate the risk-management extension planned for later checkpoints. They also make the evaluation problem more important: flexible models can overfit, so the proposed ML component must be compared with an unchanged rules-based baseline using chronological, out-of-sample tests.

3. Methods

3.1 Data sources, assets, and sample

I downloaded daily price histories for SPY, XLF, XLV, IEF, GLD, and SHY from Yahoo Finance using the yfinance Python library on October 26, 2026 (Yahoo Finance 2026; Aroussi 2026). I requested available data beginning January 1, 1993. The download retains Open, High, Low, Close, Adjusted Close, Volume, and distribution or split records when provided. I use Adjusted Close as an approximation of the value of reinvesting cash distributions and accounting for stock splits; it is not an independently audited total-return series.

The investment candidates are SPY (U.S. large-cap equity), XLF (financial equities), XLV (healthcare equities), IEF (intermediate Treasuries), and GLD (gold exposure). SHY (short-duration Treasuries) serves as the defensive allocation. All six products have at least five years of history, but GLD begins in November 2004. A single six-product panel covering all of 1999–2024 is therefore not possible without using proxies.

Ticker	First available date	Daily records
SPY	1993-01-29	8,493
XLF	1998-12-22	7,003
XLV	1998-12-22	7,003
IEF	2002-07-30	6,100
GLD	2004-11-18	5,518
SHY	2002-07-30	6,100

The common dataset spans November 18, 2004 through October 26, 2026, with 5,518 price rows and 5,517 daily returns. There were no missing adjusted prices after intersecting valid trading dates. The sample covers the 2008 financial crisis and the 2020 COVID-19 market decline, although it cannot capture the earlier dot-com crash with the full asset universe.

3.2 Return calculations and preliminary risk measures

For each asset, I calculate daily percentage log returns as $r(t) = 100 \times \ln[P(t) / P(t-1)]$, where $P(t)$ is the adjusted closing price. Simple returns are $P(t) / P(t-1) - 1$. The reported historical annualized volatility is the sample standard deviation of daily decimal log returns multiplied by $\sqrt{252}$. I calculate an adjusted-price proxy CAGR from the beginning and ending prices and the number of daily intervals, and maximum drawdown as the minimum of $P(t) / \max(P(s), s \leq t) - 1$.

The data pipeline aligns all six adjusted price histories on common dates, excludes invalid prices, and saves raw histories, processed daily returns, coverage, provenance, and risk tables. The notebook contains checks for finite returns and weights that sum to one. I also recalculated the daily returns, volatility, correlations, and drawdowns from the uploaded files before writing this report.

3.3 Trading rules and future testing plan

The proposed monthly strategy remains long-only, with no leverage, short selling, or options hedge. At each completed month-end, the five candidates must have a positive trailing 126-trading-day price return and trade above their 200-day simple moving average. The two eligible instruments with the

largest momentum scores each receive 50%; if fewer than two qualify, the unused allocation moves to SHY. The next available trading day will be used for execution in the future backtest, rather than assuming a trade at the close used to compute the signal.

Checkpoint B checks signal availability and alternative lookback windows; it does not yet simulate executed portfolios. Future testing will compare SPY buy-and-hold, the rules-based portfolio, and the same portfolio with an ML downside-risk overlay. Logistic regression and XGBoost are planned to estimate the probability of a decline of more than 3% in SPY over the next calendar month. If the estimated probability exceeds 0.60, the proposed overlay halves the current equity ETF allocation and shifts the difference to SHY. Model fitting and the 0.60 threshold must be assessed using past training and validation data only.

Later research will include walk-forward evaluation, net returns after estimated turnover and management fees, and Monte Carlo paths drawn from historical return characteristics. If block resampling is used, the same time blocks must be applied to all six asset return columns to preserve cross-asset dependence, and block lengths should be varied in sensitivity analysis. No Monte Carlo performance claim is made at this checkpoint.

4. Results and Preliminary Findings

4.1 Observed return and risk characteristics

The following statistics are for individual securities over the common period. They are not fund returns, forecasts, or results of the proposed trading strategy.

Ticker	CAGR proxy	Annual vol.	Max drawdown
SPY	10.89%	18.83%	-55.19%
XLF	6.03%	28.71%	-82.69%
XLV	10.58%	16.67%	-39.17%
IEF	3.04%	6.73%	-23.92%
GLD	10.14%	18.30%	-45.56%
SHY	1.95%	1.49%	-5.71%

XLF has the highest estimated volatility in this group (28.71%) and the deepest adjusted-price drawdown (−82.69%). SHY has much lower volatility (1.49%) but also a lower CAGR proxy (1.95%). SPY returned an estimated 10.89% per year over the sample with an 18.83% annualized standard deviation and a −55.19% maximum drawdown. These differences support measuring more than average returns when evaluating the fund.

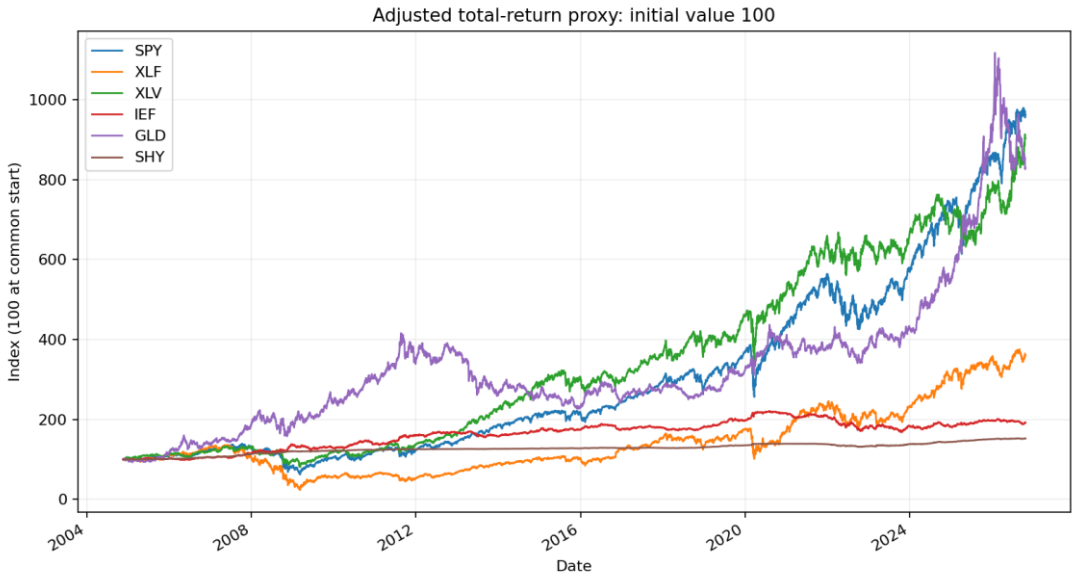


Figure 1. Adjusted-price total-return proxies rebased to 100 on November 18, 2004.

4.2 Co-movement and drawdowns

The sample daily-return correlations between SPY and XLF (0.84) and SPY and XLV (0.79) show meaningful overlap among the equity positions. SPY and IEF have a negative correlation (-0.28), while SPY and GLD are only weakly correlated (0.07) in the full sample. IEF and SHY are strongly positively correlated (0.76). These are historical associations, not stable relationships guaranteed in a future downturn.

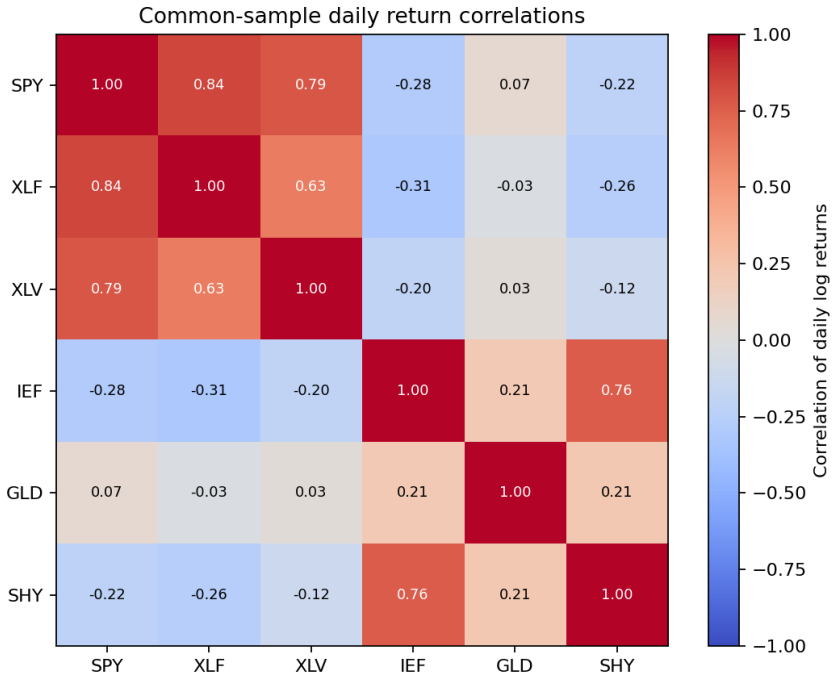


Figure 2. Correlations of adjusted-price daily log returns over the six-product common sample.

The SPY drawdown chart also captures the 2008–2009 crisis and the brief 2020 shock. The deepest drawdown in this sample occurred during the financial crisis. This gives the planned strategy a meaningful historical stress episode for testing its risk management.

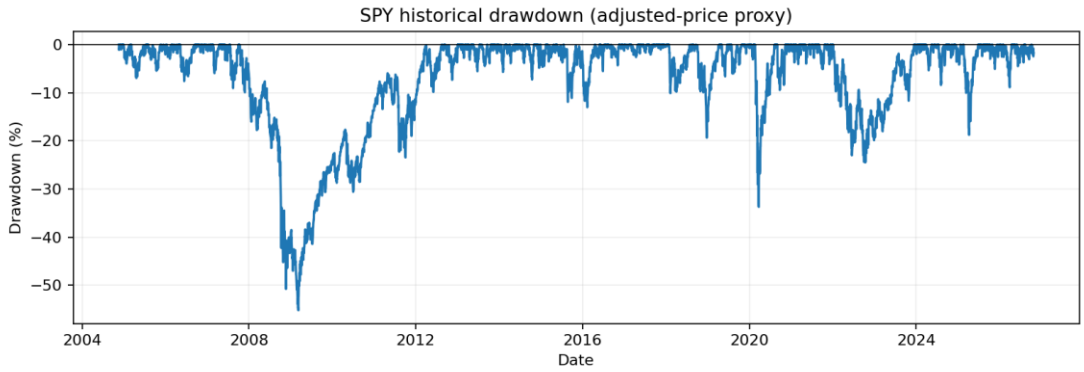


Figure 3. SPY historical drawdown based on the adjusted-price proxy.

4.3 Monthly signals and a lookback sensitivity check

I tested the availability of monthly signals using a 200-day moving average and trailing momentum windows of 63, 126, and 252 trading days. October 26, 2026 is not a completed October month-end. Therefore the formal signal preview uses completed month-end dates only, ending September 30, 2026. Signal allocations are target weights, not executed transactions or a backtest.

Momentum	Usable months	No selection	One selection	Two selections	Mean SHY target
63 days	253	13	30	210	11.07%
126 days	253	7	18	228	6.32%
252 days	251	8	23	220	7.77%

With the planned 126-day window, two securities qualified for selection in 228 of 253 completed months; one qualified in 18 months and none in 7. The corresponding average SHY target allocation was 6.32%. With a shorter 63-day window, the comparable target was 11.07%; with a 252-day window, it was 7.77% over 251 fully observable months. This is an initial sensitivity check on signal availability, not evidence that one window produces better returns. The 126-day rule is retained from Checkpoint A pending proper backtesting.

5. Conclusions, Limitations, and Next Steps

The historical data needed for the proposed six-product fund are available for approximately twenty-two years. The common sample provides multiple difficult market periods, sufficient data for exploratory risk statistics, and enough history to calculate the planned 126-day momentum and 200-day trend rules. Large risk differences and the high correlations among the equity ETFs reinforce the rationale for explicit defensive positioning.

There are important limitations. Adjusted Close is a provider-calculated total-return proxy, not a separately audited reinvestment record. The common panel cannot reproduce the full 1999–2024 span requested as an ideal example. Historical correlations and the relative performance of these securities will not necessarily persist. Most importantly, the monthly allocation preview does not show realized

fund returns, transaction costs, or whether the ML risk signal has predictive value. The latest incomplete month was excluded from formal month-end analysis.

For the next checkpoint, I will implement next-day monthly trading on the observed dataset, calculate turnover and net-of-cost returns, and compare the rules-based portfolio with SPY buy-and-hold. Then I will train and validate the two ML risk models chronologically and use market-calibrated Monte Carlo tests to examine whether apparent performance differences persist across many plausible return paths. I will report any negative or inconclusive results as well as favorable results.

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Appendix: Reproducibility and Data Notes

The repository includes the six raw ticker histories; aligned adjusted prices; daily simple and percentage log returns; a coverage table; a risk summary; a return-correlation matrix; the figures shown above; monthly signal previews and lookback sensitivity files (clearly labeled NOT_BACKTEST); source-provenance JSON; and an executed Jupyter Notebook with HTML export. The archived run uses saved raw CSVs and does not require an internet connection. Setting DOWNLOAD_FRESH=True requests a new vintage of the same public data and may change the sample and metrics.

Data checks performed: 5,518 unique observed dates, zero missing aligned adjusted prices, no weekend dates, simple and log returns agree with adjusted-price ratios, all recomputed risk estimates and pairwise correlations agree with the saved CSV summaries, and all preview monthly target weights sum to one.